# 4NF Verification

The purpose of this analysis is to evaluate our updated relational schema and determine whether the identified functional dependencies (FDs) and multivalued dependencies (MVDs) satisfy Fourth Normal Form (4NF).

We know a relation satisfies 4NF if every non-trivial MVD $X \twoheadrightarrow Y$ has a superkey $X$.

To perform our verification, we use the following properties:

- Every FD $X \rightarrow Y$ also implies MVD $X \twoheadrightarrow Y$
- An MVD is trivial if $Y \subseteq X$ or $X \cup Y$ contains all attributes in the relation
- An MVD can also describe independent sets of values, even when there is no corresponding FD

We evaluate the FDs and MVDs, disregard trivial MVDs, calculate attribute closures to test superkeys, and decompose only if a non-trivial MVD has a determinant that is not a superkey.

# FD and MVD Analysis

## 1. Users

Relation: USERS(user_id, name, email, password, date_joined)

- Primary key: {user_id}
- Key-derived functional dependency: user_id → name, email, password, date_joined
- Implied MVDs: user_id ↠ name, user_id ↠ email, user_id ↠ password, user_id ↠ date_joined
- Attribute closure: $\{user_id\}^+$ = {user_id, name, email, password, date_joined}
- Superkey: user_id (yes)

4NF Evaluation:
The MVDs implied by the identified FD satisfy the 4NF condition because user_id is a superkey and its closure contains every attribute in USERS. No additional 4NF-violating dependencies have been identified in the current analysis.

## 2. STUDY_SPOTS

Relation: STUDY_SPOTS(spot_id, name, location, type)

- Primary key: {spot_id}
- Key-derived functional dependency: spot_id → name, location, type
- Implied MVDs: spot_id ↠ name, spot_id ↠ location, spot_id ↠ type

- Attribute closure: $\{spot_id\}^+$ = {spot_id, name, location, type}
- Superkey: spot_id (yes)

4NF Evaluation:
The MVDs implied by the identified FD satisfy the 4NF condition because spot_id is a superkey and its closure contains every attribute in STUDY_SPOTS. No additional 4NF-violating dependencies have been identified in the current analysis.

## 3. REVIEWS

Relation: REVIEWS(review_id, user_id, spot_id, review_date, rating, outlet_availability, noise_level, description)

- Primary key: {review_id}
- Foreign keys:
  - user_id references USERS(user_id)
  - spot_id references STUDY_SPOTS(spot_id)
- Key-derived functional dependency: review_id → user_id, spot_id, review_date, rating, outlet_availability, noise_level, description
- Implied MVDs: review_id ↠ user_id, review_id ↠ spot_id, review_id ↠ review_date, review_id ↠ rating, review_id ↠ outlet_availability, review_id ↠ noise_level, review_id ↠ description
- Attribute closure: $\{review_id\}^+$ = {review_id, user_id, spot_id, review_date, rating, outlet_availability, noise_level, description}
- Superkey: review_id (yes)

4NF Evaluation:
The MVDs implied by the identified FD satisfy the 4NF condition because review_id is a superkey and its closure contains every attribute in REVIEWS. No additional 4NF-violating dependencies have been identified in the current analysis.

# Independent MVD Analysis

We also considered whether any of our relations contain independent multivalued attributes beyond those implied by the identified functional dependencies.

For example, in REVIEWS, a user can submit multiple reviews with different ratings. However, each rating is associated with a specific review and study spot. This does not establish user_id ↠ rating, because ratings cannot necessarily be combined independently with every study spot reviewed by that user.

Similarly, the current STUDY_SPOTS design stores one type per study spot, and our documented requirements do not identify separate independent collections of values within that relation.

Therefore, no additional non-trivial independent MVDs have been established from our current requirements. This analysis may need to be revisited if future requirements introduce independently multivalued attributes.

# Summary

| Relation | Primary Key | Identified MVD Determinant | LHS Superkey? | 4NF Violation Identified? |
|---|---|---|---|---|
| USERS | user_id | user_id | Yes | No |
| STUDY_SPOTS | spot_id | spot_id | Yes | No |
| REVIEWS | review_id | review_id | Yes | No |

# Conclusion

Based on the identified key-derived functional dependencies and the multivalued dependencies they imply, no 4NF violations have been identified in our updated relational schema. Each identified dependency has a superkey on its left-hand side, as demonstrated by the attribute closures above. Our current requirements do not establish any additional independent non-trivial MVDs that would require decomposition. Therefore, no further decomposition is necessary based on the dependencies analyzed.